

The Graph Laplacian and the Continuous Laplace Operator

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<https://github.com/zacharydorson/graph-laplacian-manifold-geometry>

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Abstract

We give the necessary background, reviewing the graph Laplacian L , the continuous Laplace operator Δ , and the Laplace-Beltrami operator, the generalization of Δ from $\mathbb{R}^n$ to Riemannian manifolds. We then discuss how one can use the graph Laplacian L as a discrete approximation of the continuous Laplace operator Δ to compute a representation of the Riemannian metric on M , which gives us a way to approximate many pieces of geometric information about M . We implement the appropriately scaled graph Laplacian, following [1], and test this method on synthetic datasets sampled from low-dimensional manifolds.

1 Introduction

We begin with a review of the central objects. The continuous Laplace operator Δ on $\mathbb{R}^n$ is the sum of the second unmixed partial derivatives

$$\Delta f = - \sum_{i=1}^n \partial_i^2 f \tag{1}$$

where ∂_i is the partial derivative with respect to x_i . Letting $S_\epsilon(x_0)$ be the sphere of radius ϵ centered at a point $x_0 \in \mathbb{R}^n$, we have another formulation of the value of Δf at x_0

$$(\Delta f)(x_0) = - \lim_{\epsilon \rightarrow 0} \frac{2n}{\epsilon^2} \left(\frac{1}{|S_\epsilon(x_0)|} \int_{S_\epsilon(x_0)} f(x) dx - f(x_0) \right) \tag{2}$$

Where $|S_\epsilon(x_0)|$ is the surface area of the sphere. That is, we find the difference between the average value of f on a sphere centered at x_0 and f at x_0 , and take the limit as the sphere's radius goes to 0.

Due to this interpretation of Δ as a measure of diffusion, the Laplace operator is indispensable in physics. For example, solutions $u(x, t)$ to the heat equation $u_t = -\alpha \Delta u(x, t)$ on some domain D model heat flow through a material (depending on D and α), so in particular if we consider stationary u , so $u_t = 0$, solutions describe equilibrium distributions of heat across the material. This operator has also found great success in more abstract fields of math, being used to study general smooth manifolds through tools like the Hodge theorem, which tells us (roughly) that each (de Rham) cohomology class of M has a unique representative as an element of $\ker \Delta_M$ with minimal L^2 energy. So, this is a sense in which the continuous Laplace

operator contains a lot of information about the geometry of the space it's defined on.

We note that when considering a general Riemannian manifold M rather than $\mathbb{R}^n$, we define the generalization of the Laplace operator, the Laplace-Beltrami operator Δ_M , in local coordinates as

$$\Delta_M f = -\frac{1}{\sqrt{|g|}} \partial_i (\sqrt{|g|} g^{ij} \partial_j f) \quad (3)$$

where $|g|$ is the determinant of the metric tensor g , g^{ij} is the ij entry of the inverse metric tensor, and ∂_i is the partial with respect to x_i . Then in the case $M = \mathbb{R}^n$ we have $g = I$, so this reduces to exactly equation (1). We write this definition for completeness, but this formulation is not necessary for any of our discussion. We have an alternate formulation $\Delta f = -\text{div}(\nabla f)$, the divergence of the gradient of f , which we will use later for a computation.

Now we will give a brief review of the graph Laplacian and some basic spectral clustering ideas. Given a graph $G = (V, E)$ with (undirected) adjacency matrix A and diagonal degree matrix $D_{ii} = \sum_{j \neq i} A_{ij}$ we define the graph Laplacian L as

$$L = D - A \quad (4)$$

Since A is symmetric and D is diagonal we immediately have that L is symmetric. We state but do not derive the identities $(Lv)_i = \sum_{j \neq i} A_{ij}(v_i - v_j)$ and $x^T Lx = \frac{1}{2} \sum_{ij} A_{ij}(x_i - x_j)^2$. The quadratic form $x^T Lx$ is clearly non-negative, so the matrix is positive semi-definite. It is easy to see that L always has a nontrivial kernel, since $(L\mathbf{1})_i = L_{ii} + \sum_{j \neq i} L_{ij} = \sum_{j \neq i} A_{ij} + \sum_{j \neq i} -A_{ij} = 0$. So L always has at least one zero eigenvalue. In particular we have one zero eigenvalue for each connected component, which may remind the reader of the 0'th cohomology group $H^0(G) \cong \mathbb{R}^c$ for c the number of connected components in G .

Let w be the eigenvector associated with the smallest nonzero eigenvalue λ of L . We call w the Fiedler vector, and it is known that this solves the (relaxed) minimal graph cut problem, where we want to partition the vertex set V into subsets V_1 and V_2 while minimizing a cost based on edges between vertices in opposite clusters. In particular for each vertex v_i we have an entry w_i , and we cluster based on the sign of this entry. To see why this works, let $f(v_i)$ be an indicator-type function on the V_i 's so that $f(v_i) = 1$ if $v_i \in V_1$ and $f(v_i) = -1$ if $v_i \in V_2$. Then an edge goes between clusters if the quantity $A_{ij}(f(v_i) - f(v_j))^2 \geq 0$. So considering f as a vector with $f_i = f(v_i)$, we can write our cost as $f^T Lf = \sum_{ij} A_{ij}(f_i - f_j)^2$. Since $\|f\|^2 = n$ the above is a Rayleigh quotient up to a constant factor. We throw out the trivial minimizer $\mathbf{1}$, which corresponds to having every vertex in the same cluster. So, that formulation gives exactly the Fiedler vector w as the best nontrivial clustering.

We will consider graphs derived from data. First we state some strong assumptions. We are interested in the case where the data is assumed to be sampled from some strongly structured embedded subspace $M \hookrightarrow \mathbb{R}^n$, as in the manifold hypothesis. That is, given some collection of datapoints $x_i \in \mathbb{R}^n$ and some symmetric positive definite measure of similarity $s(x_i, x_j)$ we start with the adjacency $A_{ij} = s(x_i, x_j)$ and then compute the graph Laplacian L as before. Now we state the central motivating question for this report: Given the analogy between the continuous Laplace operator Δ and the graph Laplacian L , can we extract the geometric information contained in Δ_M through the matrix L ? This is an attractive prospect because L is a tractable numerical object, unlike the infinite-dimensional Δ_M . Indeed, Belkin and Niyogi[2] showed in 2008 that, up to correct scaling and choice of s which will be discussed in the proceeding sections, the graph Laplacian L constructed with data x_i as above converges pointwise to the continuous Laplace-Beltrami operator Δ_M on M .

2 The Method

2.1 The Idea

The Riemannian metric g on M is an inner product on the tangent space $T_p M$ at each point $p \in M$. If $x, y \in T_p M$ we write $g_p(x, y)$. Since we think of M as being embedded in $\mathbb{R}^n$, we may view these tangent spaces as subspaces, and the metric as a weighted Euclidean dot product restricted to $T_p M$. That is, we may write $g_p(x, y) = x^T G_p y$ for a symmetric PSD matrix G_p . Our goal will be to compute the inner product $\nabla f^T G_p \nabla g$ for appropriately chosen f and g so that we can recover geometric information from G_p . To that end, we compute the Laplacian of the product $\Delta(fg)$, using the formulation $\Delta f = -\text{div}(\nabla f)$

First, the product rule for the gradient gives $\nabla(fg) = f\nabla g + g\nabla f$. Then since $(f\nabla g)_i = f(\partial_i g)$, the divergence of the first term is

$$\sum_i \partial_i [f(\partial_i g)] = f \sum_i \partial_i^2 g + \nabla f^T \cdot \nabla g.$$

In particular everything is with respect to the Riemannian metric, so we may write the inner products in terms of our matrix representation of the metric as $\nabla f^T G_p \nabla g$. But then this is exactly $-f\Delta g + \nabla f^T G_p \nabla g$, and the divergence of the second term is given by simply replacing f by g , so since the inner product is symmetric the sum simplifies to

$$-\Delta(fg) = -f\Delta g - g\Delta f + 2\nabla f^T G_p \nabla g.$$

Rearranging gives

$$\nabla f^T G_p \nabla g = \frac{1}{2}(-\Delta(fg) + f\Delta g + g\Delta f),$$

so we can write the weighted inner product $\nabla f^T G_p \nabla g$ in terms of just the Laplacian Δ .

Now let f_i be the coordinate function taking a vector x to its i 'th coordinate x_i . Then the Euclidean gradients ∇f_i are exactly the standard basis vectors e_i , so in particular the set of gradients $\{\nabla f_i\}$ spans $\mathbb{R}^n$. Then the projection of that set onto the tangent space $T_p M$ would span, so if we could compute that projection we would get a basis for $T_p M$, which is exactly what we want. Let A denote the matrix with i 'th column the projection of $e_i = \nabla f_i$ onto $T_p M$. We can compute $\nabla f_i^T G_p \nabla f_j$ for all i and j , which is exactly $(A^T A)_{ij}$, so we can compute the gram matrix of A , which has rank and span equal to A . So we have a path to compute the tangent space from the Laplace operator. Now we describe how to construct L so that it converges to Δ , allowing us to make use of the above expressions by simply replacing Δ by L . That is, we think of f and g as functions on our datapoints and replace the application of the Laplace operator with a matrix-vector multiplication Lf .

2.2 Constructing The Discrete Laplacian

To guarantee that L converges to Δ we must use an appropriate similarity metric and appropriate scaling. We will construct the adjacency matrix K as

$$K_{ij} = \exp\left(\frac{-\|x_i - x_j\|^2}{\epsilon \rho(x_i) \rho(x_j)}\right) \quad (5)$$

Where ϵ is a global bandwidth, and ρ is a bandwidth function defined on our data points. The function ρ is determined automatically by estimating the sample density around each

point. In particular, this modification, and more modifications discussed in [1] and [3], were added to account for non-uniform sampling. As we will discuss later, for all our results we uniformly sample synthetic data. However, we found that adding the variable bandwidth kernel improved our results significantly. We suspect that the relatively small datasets we generated were still non-uniform enough that performance suffered, since, for instance, if our samples from a square are low density somewhere in the interior, this is hard to distinguish from a hole with a 1-dimensional boundary.

We assemble K in three steps. Starting with an initial bandwidth function ρ_0 and a set of candidate ϵ 's, we use a heuristic to pick the best ϵ . Briefly, picking the ϵ should cause the mean of the kernel entries $\langle K \rangle = \frac{1}{N^2} \sum_{i,j} K_{ij}$ to grow like $\epsilon^{\frac{d}{2}}$ up to a constant. Then taking the logarithm of both sides gives $\log \langle K \rangle = \frac{d}{2} \log \epsilon$, so the $\log \epsilon$ -derivative of $\langle K \rangle$ should be $\frac{d}{2}$. The empirical heuristic is that this also coincides with the largest slope of the graph of $\log \langle K \rangle$ against $\log \epsilon$, so we simply numerically differentiate with respect to $\log \epsilon$ and pick the ϵ giving the largest slope. We give more details in the appendix. We use our chosen ϵ and our initial ρ to construct an initial kernel K_0 . From this we derive the sampling density estimate q_{est} and use this to compute a final bandwidth function ρ . Then with this ρ we repeat the above heuristic to choose the best ϵ with respect to our new ρ . Finally, we compute K and assemble the Laplacian. In particular, the scaled Laplacian we compute is

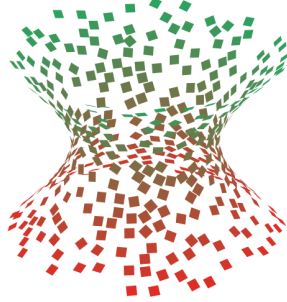
$$L = \text{diag}(\frac{1}{\rho^2}) \frac{I - D^{-1}K}{\epsilon}$$

where D is the degree matrix containing the row sums of K and the diagonal of a vector $\text{diag}(\frac{1}{\rho^2})$ is the matrix $(\text{diag}(\frac{1}{\rho^2}))_i = \frac{1}{\rho_i^2}$. So, we are computing a normalized graph Laplacian and then scaling by ϵ and the inverse of the variable bandwidth function ρ . We omit discussion of some small implementation details from [1], but we do mention one key step: after computing the final kernel K we assemble a diagonal matrix $S_{ii} = (\sum_j K_{ij})^{-\frac{1}{2}}$. This is the weight with respect to which our operator L is self-adjoint/symmetric, so we keep track of this matrix to use in our future computations.

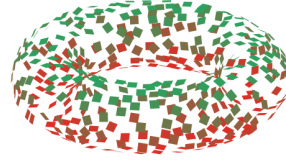
3 Results

3.1 Tangent Bundle

Now we may use the product rule formula discussion from earlier and our discrete Laplace operator L to compute a matrix A_{x_i} associated to each point x_i , where $(A_{x_i})_{ij} = \nabla f_i^T G_p \nabla f_j$. Before we do this we smooth our data using the eigenvectors of L , which correspond to eigenfunctions of Δ . These eigenfunctions are analogous to Fourier modes on the space (e.g. they are sines and cosines on the circle), so projecting our data onto the space spanned by the first n_0 eigenvectors of L is analogous to smoothing a function by removing its high-frequency components, which improves our estimations. In particular we weight this projection with respect to the sampling density matrix S from above. Then, as discussed in section 2, we may diagonalize this matrix to find a basis for the tangent space of our data at x_i . We expect to have d nonzero eigenvalues where our data is d -dimensional, with the associated eigenvectors spanning the tangent space. The remaining $n - d$ (where n is the ambient dimension, e.g. $x_i \in \mathbb{R}^n$) eigenvalues are expected to be 0, with associated eigenvectors spanning the space normal to the tangent space. Figure 1 plots a subset of the computed tangent planes for two surfaces, a torus and a hyperboloid of one sheet.



(a) Hyperboloid



(b) Torus

Figure 1: Computed tangent planes

3.2 Pointwise Dimension

We may also use these matrices to estimate the dimension of the data at each point. Assuming the eigenvalues are sorted in descending order, we find the smallest entry in the vector of forward differences $(1 - \lambda_0^i, \lambda_1^i - \lambda_0^i, \lambda_2^i - \lambda_1^i, \dots, \lambda_n^i - \lambda_{n-1}^i)$ of the eigenvalues of A_{x_i} . Since we assume that after normalization λ_j^i is either 0 or 1, this will be -1 at index d and 0 elsewhere. In practice we do not get eigenvalues that are exactly 1 or 0, but regardless we should see the largest negative value at index d . In our code we also scale the eigenvalues by their median to normalize their scale across points. Figure 2 shows a plot of the dataset with points colored based on estimated dimension.

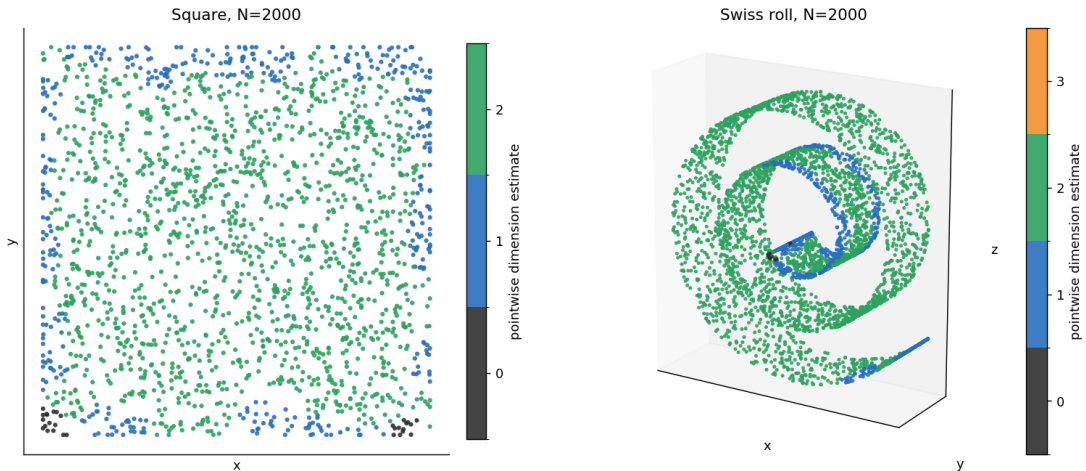


Figure 2: Pointwise dimension estimates

3.3 Added Noise

The data used for the above plots was sampled without noise, but we may wonder what happens when we are using noisier data. We sample a synthetic dataset from a unit sphere $S^2 \subset \mathbb{R}^3$ and plot the results of the above computations against a noise parameter σ . For each value of σ we add vectors with coordinates drawn independently from $N(0, \sigma^2)$ to all datapoints, find the dimension estimates and normal direction estimates, and compare them to the ground truth dimension and normals for the sphere. We see that the dimension estimate is very robust to added noise, while the normal direction degrades much faster.

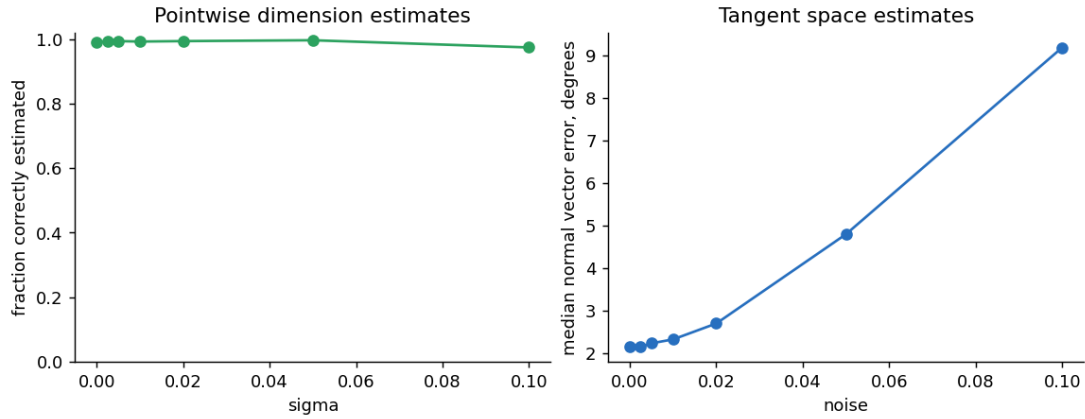


Figure 3: Estimates with added noise

Appendix A

As discussed in section 2.2, we choose the bandwidth parameter ϵ using a heuristic discussed in [4]. Here we give more details. We are considering the mean of the kernel elements, $\langle K \rangle = \frac{1}{N^2} \sum_{i,j} K_{ij}$, where $K_{ij} = \exp(-\frac{\|x_i - x_j\|^2}{\epsilon})$. This sum is exactly an approximation of the integral

$$\int_M \exp\left(-\frac{\|x_i - z\|^2}{\epsilon}\right) dV,$$

where the integral with respect to the volume form dV on M is taken over all points z . So in particular, the correct choice of ϵ should make the kernel sum a good approximation for the integral. But when ϵ is small points far from x_i have small contributions, so we may get yet another approximation by only considering a neighborhood around x_i . But a sufficiently small neighborhood around a point looks like $\mathbb{R}^d$, so we may write another approximation

$$\int_{\mathbb{R}^d} \exp\left(-\frac{\|x_i - z\|^2}{\epsilon}\right) dz.$$

But this is just a Gaussian integral, and a routine computation gives the value as $(\pi\epsilon)^{\frac{d}{2}}$. So, with the right choice of ϵ the kernel sum should scale like $(\pi\epsilon)^{\frac{d}{2}}$ (up to a constant) with respect to ϵ . But if $\epsilon \rightarrow 0$ then the kernel entries approach 0 when $i \neq j$ and 1 otherwise, so the mean should be exactly $\frac{1}{N^2} N = \frac{1}{N}$. Then when $\epsilon \rightarrow \infty$ the kernel entries all go to 1, so we get $\frac{1}{N^2} N^2 = 1$. So, there is some regime in the middle where the kernel sum scales linearly with $\log \epsilon$, and this is the regime we'd like to be in. The left figure below shows a plot of $\langle K \rangle$ against ϵ , and the right shows the fraction of correct pointwise dimension estimates against ϵ . Both have a dotted vertical line corresponding to the ϵ chosen by the heuristic. The dimension estimates were done with points sampled from a unit sphere as above, and we sampled the mean of the final kernel matrix. We see that the dimension estimates get better when the slope is largest and then mostly level out, getting a bit worse for very large ϵ .

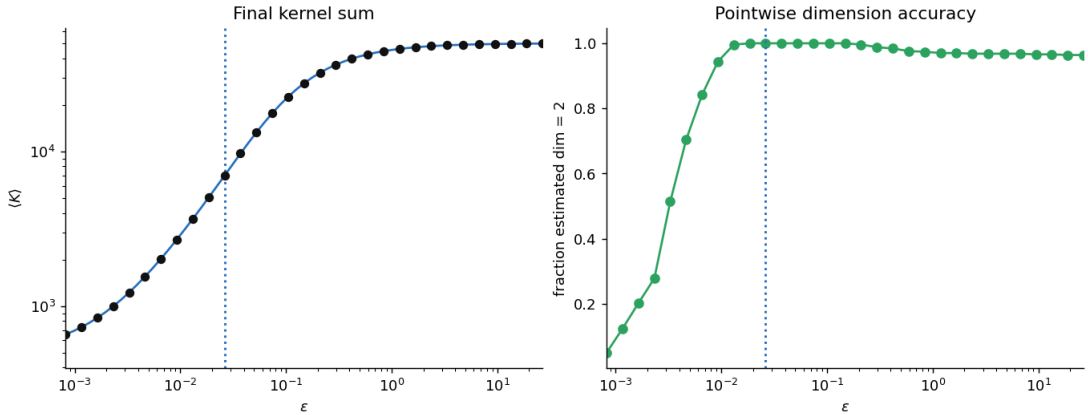


Figure 4: Kernel sum and dimension estimation against ϵ

References

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